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Game Geometric Dispersion Model with Comparative Out-of-Sample Empirical Study in Other-Regarding Preferences

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***Abstract*—The Nash equilibrium (NE) assumes that each game player (agent) is self-interested and maximizes only her own expected payoffs given opponents' strategies. Experiments have shown that players are also influenced by their opponents' payoffs. To represent such behavior, other-regarding models include the opponents' payoffs in the players' preference utilities. Existing other-regarding models are either linear in the payoff differences between players or use a single fixed nonlinear form that does not vary when the player is at an advantage or disadvantage. We propose the Game-Geometric Dispersion (G-GD) model, which displays other-regarding preferences through asymmetric nonlinear adjustments applied independently to the advantaged and disadvantaged branches, thus representing self-interest as well as four types of other-regarding behavioral preferences. The adjustment to the player's own payoff is based on Geometric Dispersion (GD), a nonlinear measure of the dispersion among all players' payoffs. G-GD is parsimonious, using only two parameters, and its transformation of the payoff matrix adds negligible computational cost to solving for the NE. Using 80 observed choice distributions in three datasets from four published experimental studies of actual human behavior, we compare G-GD, NE, and three leading other-regarding models. In each of the eight repetitions, model parameters are estimated on a randomly selected in-sample set of the data and predictions are tested on the remaining out-of-sample set. G-GD attains the highest mean out-of-sample log-likelihood in all three datasets, and its advantage over its closest competitor models is statistically significant at the 5% level. The estimated preference types differ by game category, suggesting that other-regarding behavior depends on the strategic context.**

***Index Terms*— Game, agent, player, experimental studies, matching pennies, Kreps, rock-paper-scissors, other-regarding preferences, fairness, inequity aversion, social welfare, one-shot games, non-cooperative, out-of-sample.**

I. INTRODUCTION

GAME theory has a significant impact on systems engineering, design, and analysis for applications in economics, management, and government. When humans compete or collaborate for resources and survival, game-theory strategies become vital. The Nash equilibrium (NE) assumes fully rational, self-interested, risk-neutral players, but numerous experiments show systematic deviations from NE predictions when payoffs change or asymmetries are introduced ([1], [2], [3]). Despite this, the NE remains central in solving for steady states in many critical real-world applications, such as choices in electricity markets [4], [5], production decisions [6], routing delivery drivers [7], social politics and pollution control [8], rumor-mongering in online social networks [9], or allocation of medical resources in a pandemic [10]. Across these applications, accurately predicting the behavioral equilibrium is fundamental to system functionality.

Behavioral discrepancies with NE cannot be explained solely by noise, risk aversion, or bounded rationality ([11], [12], and [13]). Instead, subjects often display other-regarding preferences (fairness, altruism, reciprocity, inequity aversion) that affect choices in games ([14], [15], [16]). We use the term *other-regarding preferences* to reflect that each player's preferences or utility is not only based on their own self-interest but also depends on the payoffs of other players, such that the player's goal may be to help or penalize opponents' interests in the preference utility; and this is the theme of this paper. Economic theories have traditionally emphasized pure self-interest, overlooking the behavioral and economic implications of other-regarding preferences. For example, one-shot games (i.e., simultaneous move and non-cooperative) are particularly relevant in elections, military strategy, auctions, legal disputes, politics, and business ([2], [17], [14]). The most prominent other-regarding models include the inequity aversion utility of Fehr & Schmidt (FS) [18], the fairness model of Charness & Rabin (CR) [19], and the nonlinear utility of Bolton & Ockenfels (BO) [20]. Yet these models face limitations: Boylan & Grant [12] demonstrate that linear inequity aversion is inconsistent with behavior in asymmetric matching pennies, and Engelmann & Strobel ([21], [15]), Bergh [22], and Cooper et al. [23] all further document weaknesses in the predictive power of all three models.

Another widely studied approach is quantal response equilibrium (QRE) by McKelvey et al. [24], which models stochastic choice through payoff randomization and logit-based probability weights. Although there is a lot of support for the use of QRE in comparison benchmarks in place of NE ([17], [3], [25]), its falsifiability is limited since the randomization distribution can be tuned to match behavior in any context [26]. Alternative bounded rationality approaches, including level-k and cognitive hierarchy (CH) models [27], also require strong assumptions about beliefs and initializations, which introduce free parameters, increase complexity, risk overfitting, and undermine cross-game performance ([28], [29], [25]).

In the recent work in [30], an extension of Geometric Dispersion Theory (GDT) [31] is used to explain decisions under ambiguity and uncertainty better than alternative models in out-of-sample behavioral studies. Yukalov et al. [32] show

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that decisions of players in games can also be modeled as decisions under uncertainty where the probabilities of the opponent's choices are unknown. In the game-theoretic approach in [33], expert opinions determine these probabilities while players change their choices based on their pre-determined type. On the other hand, in this paper, we propose a model that predicts the player's behavior in games based on their own and opponents' payoffs, a more general and practical approach.

In this paper, we develop the Game Geometric Dispersion (G-GD) model based on the GDT framework. GDT was developed as a normative model to explain decision behavior better than leading models, both under risk and for multi-criteria [37] [38]. One component of GDT, called Geometric Dispersion (GD), measures the deviation (or dispersion) from the expected value. In this paper, we develop the G-GD model for other-regarding utility for game problems using GD. While GDT generalizes expected utility for the risk problem, G-GD generalizes the NE for the game problem.

G-GD explains behavior in games by using nonlinear other-regarding preferences (i.e., preferences influenced by others' payoffs as well as the player's own) to transform the payoff matrix, which is then solved by NE. In G-GD we use the GD measure to represent nonlinearity in payoffs with independent utility adjustments depending on whether the player is advantaged or disadvantaged relative to the opponents' payoffs, while maintaining parsimony with only two parameters similar to the FS and CR models. Using data from four experimental studies in the literature with a total of 80 observed probabilities of choices in games aggregated from thousands of subjects' choices, we show that the nonlinear utility enables G-GD to more accurately represent observed behavior than NE and the three other-regarding models tested, in out-of-sample testing across the three datasets studied. In addition, we explore the properties of G-GD and show that it represents behavioral other-regarding preferences, diminishing returns in payoffs, and the inclusion of non-Nash strategies in equilibrium solutions.

This paper is organized as follows. In Section II, the models of game theory relevant to this paper are discussed. In Section III, we present three well-known game applications for which human decision-making behavior deviates from the theoretical equilibrium solution of NE. Section IV reviews the GDT for risk problems. Section V discusses the proposed G-GD model using GDT. Section VI presents experimental data, the parametrization of game theory models, and the results of the experimental analysis comparing G-GD to other models. Section VII presents the discussion, and Section VIII contains the conclusion. Details of experiments are shown in the online supplement [34].

II. Modeling Decisions in Game Theory

Games can be classified by symmetry (of the payoff matrix for all players), player interaction (e.g., one-shot or cooperative), number of players, or other characteristics (e.g. strategic solutions such as pure equilibria, etc.). In this paper, we develop a general theory, but for applications, we focus on games that are normal-form (i.e., with a fully defined payoff matrix) and one-shot (simultaneous-move or non-cooperative) to model the individual player's best response behavior.

Models with free parameters and assumptions that are not transferable across applications can rationalize any observed distribution of behavior and are non-falsifiable or difficult to reject on any particular dataset. Haile et al. [26] formally demonstrate that QRE, absent additional restrictions on the error structure, imposes no falsifiable content on observed behavior, since the context-dependent error distribution can be chosen after the fact to rationalize any choice pattern. The same free-parameter concern applies to level-k and cognitive hierarchy models, whose predictions depend on an assumed Level-0 anchor and number of reasoning levels that are not disciplined by theory [25]. As a result, we restrict the model comparison only to the models (NE, FS, CR, BO, and G-GD) whose preferences are based on behavioral properties of the player's utility and payoff structure of the game rather than on auxiliary assumptions, and for the remainder of this paper, we only address these models, which are related to the contribution of this paper. In addition, learning models describe how behavior changes with repetition [27], whereas G-GD and other-regarding models explain why choices depart from the Nash prediction in a single encounter. Therefore, our claim is to test whether G-GD's nonlinear other-regarding preferences explain behavior in games better than the linear and fixed-nonlinear alternatives tested, not that preference-based explanation supersedes learning where learning is possible. In the following, we present NE first, which assumes the players are self-interested, then go on to explain (FS, CR, BO) in which the player's utility also considers the payoffs of opponents.

A. Nash Equilibrium (NE) for Self-Interested Preferences

The NE solution assumes players are self-interested and only maximize their expected value (EV) of payoffs. The NE point is reached when no player will change their selected strategy (or alternative), even if the other players' strategies are made known [1]. To solve for the NE, the optimization finds a strategy selection in which each player's strategy maximizes her own expected payoff given the other players' strategies (see [1]; [35]; [36]). The equilibrium solution may be a pure strategy or a mixed equilibrium of probabilities of strategies.

Consider an n-player game, where a player, k, where k= 1, 2, ..., n, selects a strategy, s_k, where player k's strategies are numbered such that s_k=1,…, S_k, where k=1,…,n. A strategy selection, s_k, for all k, results in a state, **s**, in the set of all possible player selections, at which the payoffs earned by each player are ($y_1, \ldots, y_n$). The game is defined such that there is a set of payoffs ($y_1, \ldots, y_n$) corresponding to each state, **s**, identified by each player's strategy selection, i.e., **s** = [s_1, …, s_k, …, s_n]**.** In this paper, in order to consider alternative utility options, each set of payoffs in state **s**, ($y_1, \ldots, y_n$), is replaced by a utility-transformed payoffs, ($u_1(y_1, \ldots, y_n), \ldots, u_n(y_1, \ldots, y_n)$), or ($y^*_1, \ldots, y^*_n$), where the player's utility, u_k, can be a function of their own and opponents' payoffs in that state. The equilibrium is then found by using ($y^*_1, \ldots, y^*_n$) in place of the original payoffs, ($y_1,\ldots,y_n$), in the EV expressions in the NE optimization [19]. Using this notation, the player's utility in NE is dependent on her own payoff only, i.e., for player k, $u_k(y_1, \ldots, y_n) = y_k$ for all k=1,..,n.

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B. Nash Equilibrium with Other-Regarding Preferences

In this paper, we consider methods whose players have a utility that is a function of their own and opponents' payoffs (y_1, …, y_n) in the current state, or $u_k(y_1,\ldots,y_n)$; we call these *other-regarding preferences*. Under such a utility, players are not purely self-interested, and behavioral deviations from NE occur. We follow the solution strategy of Section II.A throughout the paper using alternative utility models to transform the payoff matrix.

The behavior displayed by models of other-regarding utilities varies based on the marginal utility of the Decision Maker's (DM) preferences. The marginal utility with respect to the DM's own interest is generally assumed to be positive, but there are exceptions. Altruism (or unconditional kindness) is characterized by an increase in the DM's utility with an increase in the other player's payoff, whereas Envy (or Spitefulness) is when the DM increases their utility due to a decrease in others' payoffs. These preferences can also be conditional, such as Inequity Aversion (or conditional Altruism), where the DM will demonstrate altruism in a less equitable situation, whereas a more equitable situation will result in envious behavior. The following are models of other-regarding preferences. Intention-based reciprocity effects are omitted in this paper's presentation since these are irrelevant for one-shot non-cooperative games.

1) Fehr & Schmidt fairness model (FS)

Fehr and Schmidt [18] develop a model of fairness (inequity aversion) (FS), which is a simple linear model that incorporates other-regarding preferences by balancing the player's own payoff with other players' payoffs. The model penalizes the EV by the average advantage or disadvantage (the absolute difference of the player's payoff from that of other players). In both cases, the players are interested in achieving egalitarian fairness (i.e., all payoffs are the same).

2) Charness & Rabin social welfare model (CR)

Charness and Rabin [19] introduced a social welfare utility model (CR), which is a combination of reciprocity and other-regarding behavior. Given this paper's focus on the non-cooperative one-shot game, we present only the other-regarding preferences component of the CR model, and reciprocity effects are excluded from this presentation. The CR utility is a convex combination of the player's own payoff and the social welfare term. The social welfare term is also a convex combination of the lowest and total payoffs for the entire group, i.e., the CR preferences combine motives of selfishness, Maximin, and efficiency. The CR model assumes homogeneous other-regarding preferences in the population. The CR model can represent various other-regarding preference behaviors such as social welfare, inequity aversion, altruism, and envy, depending on the marginal utility of the player's own payoffs and the opponent's payoffs.

3) Bolton-Ockenfels inequity aversion model (BO)

Bolton and Ockenfels [20] propose a general form of non-linear other-regarding preferences (BO) in which preferences are affected by the proportion of payoffs allocated to each player to represent inequity aversion. Like the CR preference utility, the BO preferences consider equity among the whole group rather than equity between pairs of players as in the FS utility. However, in two-player games, these differences become less pronounced. The BO model achieves inequity aversion using a nonlinear concave function; however, that makes the BO model unable to represent other behaviors that CR does, such as altruism and envy.

III. Experimental Behavioral Results and Paradoxes that Contradict the Nash Equilibrium

A. Empirical Results of Asymmetric Matching Pennies

Behavior in the matching pennies game in its original form follows the prediction of the NE well; however, when asymmetry in the payoffs is introduced, experimental behavior is found to deviate from the NE. The payoff matrix of the original matching pennies game and two asymmetric variations are shown in Table 3.1. In all three of these variations, the only change is the payoffs of the row player (R), while no changes are made to the payoffs of the column player (C). As a result, NE predicts that R chooses between A' or B' with 50% probability in all variations due to payoffs of C held constant. However, the percentage of choices reported in experiments from [2], as shown in the tables, tells a different story. The asymmetric payoffs for R seem to be at play here, given that the expected payoff is considerably higher in Table 3.1(b) than in Table 3.1(c). On the other hand, the behavior of C in experiments is not in contradiction with NE, as NE also predicts C's choice percentage changes due to the variation in payoffs for R.

Table 3.1: Payoff matrices of matching pennies game variations : (a) is symmetric, while (b) and (c) are asymmetric. Experimental choice percentages reported in [2] are shown next to each strategy.

(a) Symmetric

		Column Player (C)	
		A (48%)	B (52%)
Row Player (R)	A' (48%)	**80**, 40	40,80
	B' (52%)	40,80	80, 40

(b) Asymmetric 1

		Column Player (C)	
		A (16%)	B (84%)
Row Player (R)	A' (96%)	**320**, 40	40,80
	B' (4%)	40,80	80, 40

(c) Asymmetric 2

		Column Player (C)	
		A (80%)	B (20%)
Row Player (R)	A' (8%)	**44**, 40	40,80
	B' (92%)	40,80	80, 40

B. Empirical Results of Kreps: Non-Zero-Sum Game with a Safe Non-Nash Option

In the Kreps game payoff matrix (Table 3.2), row player has two options (A' and B') and the column player has four options (A, B, C, and D) that are ordered from best to worst in the top row and vice versa in the bottom. There are two pure Nash solutions, the choice combinations A' with A and B' with D in Table 3.2. In addition, a mixed equilibrium point exists that includes both choices for the row player and the choices A and B for the column player. The column player's choice of C is not part of any Nash solution; however, this choice for column is considered a safe one because it guarantees a gain regardless of the row player's choice. This choice is made by 68% of

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experimental subjects as reported in [2]; however, this solution is not predicted by NE.

Table 3.2: Payoff matrix in the Kreps game. Typical experimental percentage choices shown as reported in [2].

		Column Player							
		A (26%)		B (8%)		C (68%)		D (0%)	
Row Player	A' (68%)	200	50	0	45	10	30	20	-250
	B' (32%)	0	-250	10	-100	30	30	50	40

C. Empirical Results of Rock, Paper, Scissors: A Zero-Sum or Constant-Sum Game

The classic game of rock, paper, scissors is a popular example of zero-sum (or constant-sum) games. In this game, one player wins while the other loses. A game theory presentation of this game was used by Kulpmann et al. [29] is represented in Table 3.3. In this example, 0 represents a loss, 1 a win, and 0.5 is a tie. While the only NE solution to this game is a uniform randomization over the three strategies, the experimental choice frequencies reported in [29] are [37%, 32%, 31%], departing from the 1/3 prediction. This indicates a systematic bias that randomization alone does not explain.

Table 3.3: Payoff matrix in the rock, paper, scissors game as used in experiments of [29].

	Rock		Paper		Scissors	
Rock	0.5	0.5	1	0	0	1
Paper	0	1	0.5	0.5	1	0
Scissors	1	0	0	1	0.5	0.5

IV. Geometric Dispersion Theory (GDT) for the Nonlinear Multi-Criteria Decision-Making Problem

To develop the n-player other-regarding preference utility using nonlinear deviation, we rely on geometric dispersion theory (GDT) by Malakooti [37][1], a normative model of decision behavior under risk [37], multi-criteria [38], and ambiguity [30]. GDT has shown superior out-of-sample predictions against prospect theory and expected utility theory on classical risk experiments [37] [31] [30], and Malakooti, Komaki, & Al-Najjar [31] find that its geometric dispersion measures outperform classical measures, such as standard deviation used in mean-variance analysis [39], mean-absolute deviation, and semi-deviations. This section summarizes GDT for the multi-criteria problem, in which a single player (one DM) evaluates m alternatives on n criteria. Section V applies it to games with n players as part of G-GD.

A. Notations of the Multi-Criteria Problem

Consider a decision problem with multiple criteria, such that the DM evaluates alternatives over n criteria or objectives. An alternative, Y, can be represented as a set of values for the objectives or criteria, j, in the form: $(y_1,\ldots,y_n)$, where y_j denotes the value given to the jth criterion in alternative Y. Without loss of generalization, we assume all y_j are normalized such that $0 < y_j \leq 1$ for all j[2]. The DM may evaluate criteria differently, where a weight of importance, w_j, is assigned to each criterion j, for $j=1,2,\ldots,n$, in the DM's utility where $0 \leq w_j \leq 1$, such that the sum of weights for all j is 1. The DM evaluates and ranks alternatives Y with a utility, U, using the DM's set of weights, w_j, for $j=1,2,\ldots,n$. We use $\succ$ to present strict preference, $\succcurlyeq$ to present weak preference, and $\sim$ to present indifference in preference. Given alternatives Y_q, Y_r, $Y_s \in \mathbf{Y}$, a real-valued utility function, $U(Y)$ can represent the DM's preferences such that if $Y_q \succcurlyeq Y_r$, then $U(Y_q) \geq U(Y_r)$, and vice versa; see [40].

B. Geometric Dispersion for Nonlinear Multi-Criteria Problem

The linear value (expected value), denoted by EV, as shown in (4.1), represents the weighted average of all criteria in the alternative. The geometric dispersion (GD) introduced by Malakooti in [37] is a measure of deviation of the criteria and is given by (4.2). GD is the difference between the weighted arithmetic mean (EV) and the weighted geometric mean. With respect to the criterion, y_j, GD is a convex function. However, with respect to the weights, w_j, GD is a concave function. In addition, GD is always non-negative, i.e., $GD \geq 0$, while $GD = 0$ only when all y_j are equal. A distinguishing property of GD is that it is nonlinear in both payoffs and weights and cannot be transformed into a linear or additive function by logarithmic transformations [31].

$$EV = \sum_{j=1}^{n} w_j y_j \tag{4.1}$$

$$GD = EV - \prod_{j=1}^{n} y_j^{w_j} \tag{4.2}$$

C. The Basic GDT model

The Basic GDT model is a linear additive model that combines the two components of GDT, as shown in Eq. (4.3).

$$\text{Basic One Parameter GDT:} \quad U = EV + z\,GD \tag{4.3}$$

In (4.3), when $-1 < z \leq 0$, U is concave and the parameter z represents the extent of concavity; as z approaches -1, GDT becomes more concave. This model is increasing in payoffs and complies with the non-dominance (efficiency) principle for this range of z, as proven in [37]. On the other hand, when $0 \leq z < 1$ in (4.3), U is convex, and the parameter z represents the extent of convexity; as z approaches 1, GDT becomes more convex.[3] To simplify, one can assume equal weights, or $w_{hj} = 1/n$, and then assess the DM's z parameter.

While this utility differs from that used for game players in this paper, we demonstrate in the next section that the nonlinear GD deviation measure is instrumental to each player's other-regarding utility.

V. Game-Geometric Dispersion Model (G-GD)

A. Motivation and Behavioral Justification

We propose the G-GD model to explain and predict game behavior more effectively than NE and its generalizations of other-regarding preferences, such as FS, CR, and BO. In this section, we present the motivation and behavioral justifications for G-GD. Goeree & Holt [2] show that in asymmetric one-shot treasure games, players' behavior departs from the NE in a systematic way based on payoffs rather than due to errors,

[1] GDT for risk is based on the earlier idea of Multiplicative Z-Utility Theory for solving multi-criteria problems by Malakooti [38], Ch. 2.5, p. 93.

[2] Although $0 < y_j \leq 1$ is assumed, the effect is no more than that of normalization of payoffs which is commonplace in many utility applications.

[3] While $z>0$ in (4.3) results in U that is not increasing in payoffs, Malakooti [37] and Malakooti, Komaki, & Al-Najjar [31] provide a two parameter GDT model, that complies with non-dominancy principle for both convex and concave.

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noise, or limited rationality. The NE is meant to be used theoretically rather than prescriptively since it may not yield optimal or Pareto-efficient outcomes. In fact, some equilibrium points may even be unstable. Models of other-regarding preferences like BO, FS, and CR explain behavior in dictator, ultimatum, gift-exchange, and trust games, but face limits. BO is sensitive to payoff shifts, and inequity aversion appears tied to individual comparisons rather than group outcomes, as in both BO and CR([14], [41]). Experimental work comparing FS and BO by Engelmann & Strobel finds that FS's advantage over BO depends on the setting [21], and that neither model explains important patterns in their data [42] [15].

In this paper, G-GD generalizes linear other-regarding preferences within the NE framework using utility-transformed payoffs as laid out in Section II.B. G-GD uses Geometric Dispersion to measure nonlinear preferences by using only two parameters. Unlike FS and CR, whose adjustment term is linear in the payoff deviation, G-GD's adjustment term is nonlinear in that deviation. Unlike BO, which applies a single fixed concave function whether the player is ahead or behind, G-GD gives the advantaged and disadvantaged domains independent nonlinear responses (formalized in Section V.C), which allows it to represent altruism and envy in addition to inequity aversion. This is achieved with only two added parameters, the same number as FS and CR, and with no added assumptions compared to alternative solution methods such as QRE, level-k, and CH. We detail these properties next.

1) Other-regarding vs. self-interested preferences

As shown in Section III.A, in the asymmetric matching pennies, players do not behave as purely self-interested. The opponent's mixed strategy accounts for the row player's payoff as well, implying other-regarding preferences. G-GD preferences explain the systematic way that behavior varies when payoffs change asymmetrically, as illustrated in the example of Section III.A. In addition, the transformation of the payoffs results in a shift of the equilibrium, allowing non-Nash strategies to be part of the solution as in the Kreps game of Section III.B, where G-GD results in a relocation of the equilibrium itself rather than assigning it a choice probability of zero as in NE.

2) Linearity vs. non-linearity of the preferences in payoffs

Typical game-theoretic treatments assume linear utilities in payoffs, unlike decisions under risk, where diminishing marginal utility is standard practice. In asymmetric payoff shifts, large payoff increases produce smaller probability changes compared to equivalent decreases, suggesting nonlinear sensitivity. Concave utility can capture diminishing marginal utility and near-dominance effects in low-payoff cases. For example, in the asymmetric game of Table 3.1b, the payoff is increased by 4 times to shift the player's probability of choosing A' from 50% up to 96%, whereas in the asymmetric game of Table 3.1c, the payoff is roughly halved to reduce the choice probabilities by a comparable amount. In G-GD, non-linearity can capture the change in the marginal utility that the BO model can represent, while still representing the full range of preference behaviors that the linear CR model can, but BO cannot.

3) Advantaged vs. disadvantaged behavioral preferences

Other-regarding models assume players maximize their payoff and avoid being either too far ahead or too far behind others' payoffs. FS and CR address this using piecewise linear utilities, but each side (advantaged vs. disadvantaged) has fixed marginal effects. These models struggle to account for payoff shifts in asymmetric games [2] [3]. On the other hand, BO uses a fixed nonlinear function that can represent only inequity aversion, not envy or altruism. G-GD varies the utility for advantaged and disadvantaged cases and can therefore capture the full range of behavior, with the added benefit that its nonlinearity reflects varying marginal utility effects.

B. Linear Other-Regarding Preferences

The FS and CR utilities use a utility for player k, u_k, that is a linear function of the payoffs in each state **s**, $(y_1,\ldots,y_n)$, to improve predictions over NE. First, we present the linear other-regarding functions.

In n-player games, the FS utility is equivalent to a convex combination of the player's own payoff and the mean of the absolute deviations of the player's own payoff from the payoffs of all other players. In contrast, the CR utility is equivalent to a convex combination of the player's own payoff, the sum of all players' payoffs, and the minimum payoff. Therefore, both FS and CR adjust their utility by a term that measures the deviation between the player's own payoff and the opponents' payoffs. For simplicity, we present these for two-player games with a row player (R) and a column player (C), where the payoffs earned by each player in the current state based on the strategy selection of both row and column players are (y_R, y_C). For 2-player games, the linear functions for the FS and CR utilities for row player reduce to the same form shown in (5.1), which is a convex combination of R's payoff, y_R, and C's payoff, y_C, with different coefficients when R is ahead ($y_R \geq y_C$) or behind ($y_R < y_C$).

$$u_R(y_R, y_C) = \begin{cases} (1-\sigma)y_R + \sigma y_C, & \text{if } y_R < y_C \\ (1-\rho)y_R + \rho y_C, & \text{if } y_R \geq y_C \end{cases} \qquad (5.1)$$

The equation in (5.1) is re-written in (5.2) as a maximization of the player's own payoff, y_R, with an adjustment term representing the absolute difference of the players' payoffs (i.e. $|y_C - y_R| = (y_C - y_R)$ when $y_R < y_C$ or $(y_R - y_C)$ when $y_R \geq y_C$).

$$u_R(y_R, y_C) = \begin{cases} y_R + \sigma(y_C - y_R), & \text{if } y_R < y_C \\ y_R - \rho(y_R - y_C), & \text{if } y_R \geq y_C \end{cases} \qquad (5.2)$$

In the general form of the linear other-regarding utility of (5.2), the adjustment term penalizes the utility when the player has an advantage over the other player or boosts the utility when the player is at a disadvantage compared to the other player. Pure self-interested players will have $\sigma = \rho = 0$, in which case the equilibrium will default to the standard NE. On the other hand, values of σ and ρ other than zero will allow other-regarding preferences, such as altruistic preferences or inequity aversion

C. Game-Geometric Dispersion Nonlinear Other-Regarding Preferences for n Players

We propose G-GD as a nonlinear generalization of Nash and linear other-regarding utility to improve predictions of observed decision behavior over other models. In the n-player context, the set of n payoffs earned by each player in a state **s**, is $(y_1,\ldots,y_n)$, and the G-GD utility of player, k, in state **s** is $u_k(y_1,\ldots,y_n)$ where k=1,…,n. The n-player G-GD generalizes the

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linear other-regarding utility in (5.2) using the non-linear GD deviation measure in (4.2). Specifically in G-GD, the linear difference of the payoffs, y_R and y_C, i.e., (y_R - y_C) or ($y_C - y_R$), in the second term of (5.2) is replaced by the nonlinear deviation measure, *GD* (based on (4.2) in Section IV.B) such that the set of payoffs ($y_1,\ldots,y_n$) in state **s** maps to the alternative Y with n criteria. We use the payoff index i = 1, …, n instead of the criteria index j in Section IV to distinguish between the notations for games used in G-GD of this paper and for multicriteria decisions as used in GDT (Section IV.B). Then, w_i for i=1, …, n, represent the weights of importance for each criterion in the evaluation of GD. Since each player can use different weights in their utility, or w_{ki} for i=1, …, n for the kth player, the *EV* and *GD* for player k are shown as EV_k in (5.3) and GD_k in (5.4).

$$EV_k = \sum_{i=1}^{n} w_{ki} y_i \tag{5.3}$$

$$GD_k = EV_k - \prod_{i=1}^{n} y_i^{w_{ki}} \tag{5.4}$$

GD_k, as a dispersion measure, represents the non-linear deviation (or dispersion) among the set of payoffs ($y_1,\ldots,y_n$), in state ***s,*** whereas w_{ki} for i=1,…,n, is player k's weight of importance for player i's payoff, y_i,. Likewise, EV_k is player k's weighted average of all players' payoffs. Note that $0 \le w_{ki} \le 1$ for i=1,…,n and $\sum_{i=1}^{n} w_{ki} = 1$ for all k, with the assumption that $0 < y_i \le 1$ for all i. This results in G-GD as the utility for player k, as presented in (5.5) using EV_k in (5.3), GD_k in (5.4).[4]

$$u_k(y_1,\ldots,y_n) = \begin{cases} y_k + z_{k,d} GD_k, & \text{if } y_k < EV_k \\ y_k - z_{k,a} GD_k, & \text{if } y_k \ge EV_k \end{cases} \quad \text{(n-player G-GD)} \tag{5.5}$$

where simplifications: $z_{k,a} = z_a$, $z_{k,d} = z_d$, $w_{ki} = 1/n$, result in $GD_k = GD$, $EV_k = EV$ for all $i=1,\ldots,n$, $k=1,\ldots,n$.

The piecewise condition in (5.5) identifies two branches of behavior for when the player's payoff is 'good' (above the group average) or 'bad' (below the group average). While the first term, y_k, that is shared between the two branches of the piecewise function represents the self-interest or maximization of the player's own payoff, the second adjustment term varies the player's utility based on their advantaged/disadvantaged position. Like the parameters σ and ρ of FS and CR, the parameter $z_{k,d}$ represents player k's disadvantaged coefficient (when y_k is less than the weighted average, EV_k), while the parameter $z_{k,a}$ represents player k's advantaged coefficient (when $y_k \ge EV_k$).[5]

The following properties of (5.5) follow from GD_k, and the last two cannot be produced by a linear deviation term.

- Setting $z_{k,a} = z_{k,d} = 0$ gives $u_k = y_k$ in every state, so the transformed matrix is identical to the original, and the equilibrium is the standard NE. Alternatively, setting $w_{kk}=1$ and $w_{ki} = 0$ for all $i \ne k$, also leads to the special case of NE.[6]
- Since $GD_k \ge 0$, with equality only when all payoffs in state ***s*** are equal (Section IV.B), positive $z_{k,a}$ and $z_{k,d}$ represent socially concerned behavior in which the player sacrifices utility when ahead and gains utility when behind. Other sign combinations represent other social behaviors, as detailed in Section V.E.
- GD_k is convex in the payoffs, so the adjustment grows more than proportionally as the payoffs in a state disperse, whereas the deviation term used by FS and CR in (5.2) is linear in the payoff differences and grows proportionally. GD_k is also concave in weights.
- GD_k responds to relative rather than absolute differences. Adding a constant to every payoff in a state strictly decreases GD_k, since the weighted geometric mean rises by more than the constant while the weighted arithmetic mean rises by exactly the constant. The deviation term in (5.2) is unchanged by such a shift.

The parameter magnitudes also affect the shape of G-GD: for large values relative to GD_k in a state, u_k may not be monotone in the player's own payoff.

Experimentally, we found that varying w_{ki} had less explanatory power than z_a and z_d.[7] In practice, we suggest using the default values of equal weights for all players (i.e. $w_{ki} = 1/n$ for all k and i) to retain parsimony. When using equal weights, EV and GD are the same for all players, and therefore both become objective measures of the mean and dispersion, independent of the player.

The new G-GD equilibrium uses a standard NE optimization[8] of the player's best response using the transformed payoff matrix. This allows G-GD to explain the shift in the equilibrium due to asymmetric payoffs, as in the games of [2].

D. Graphical Illustration of G-GD for Two Players

In the two-player game, a row player (R) and a column player (C), make choices out of the different pure strategy sets that are available to them. At each pure strategy selection for both players, we have a pair of payoffs, (y_R, y_C), that the players earn in that state. For the 2-player game, the G-GD utility for row player is presented in (5.6) with EV_R as in (5.7) and GD_R as in (5.8). Note that when $w_{RC} = w_{RR} = 0.5$, the piecewise function condition $y_R < EV_R$ becomes equivalent to $y_R < y_C$, and $y_R \ge EV_R$ becomes equivalent to $y_R \ge y_C$. Also, with default weights, all players have the same EV and GD in that state.

$$u_R(y_R, y_C) = \begin{cases} y_R + z_{R,d} GD_R, & \text{if } y_R < EV_R \\ y_R - z_{R,a} GD_R, & \text{if } y_R \ge EV_R \end{cases} \quad \text{(G-GD)} \tag{5.6}$$

[4] Note that G-GD in (5.5) does not carry over the base term of EV from the Basic GDT model in (4.3), and instead follows the utility of (5.2), where the player maximizes their own payoff as the base term.

[5] Although the condition is written against the weighted group mean EV_k, it is equivalent to a direct comparison with opponents: writing $EV_k = w_{kk} \cdot y_k + (1 - w_{kk}) \cdot \bar{y}_{-k}$, where $\bar{y}_{-k}$ is the weighted average of all the other players' payoffs other than k, the condition $y_k \ge EV_k$ reduces to $y_k \ge \bar{y}_{-k}$ after the common factor $(1 - w_{kk})$ is canceled. The player's own weight has no effect on which branch applies. The advantaged and disadvantaged branches therefore retain the same interpretation for any n as in the two-player case, namely whether the player is ahead of or behind the typical opponent.

[6] Increasing a player's own weight interpolates toward NE: as w_{kk} approaches 1, EV_k approaches y_k and GD_k approaches zero, so the adjustment term vanishes. Which branch applies is unaffected by w_{kk}, as shown in footnote 5.

[7] Allowing the weight to vary freely alongside z_a and z_d in a game-by-game best fit over 17 two-player games from [47] [2] [3], the unconstrained fit converged to $w \approx 0.5$ in 15 games and gave only a small improvement in the other two. While this is not a formal out-of-sample comparison, it is consistent with equal weights as a default.

[8] The Nash theorem is found in [1], however, for the purposes of this paper, we use the NE algorithm and MATLAB code by Chatterjee [36] [50] which finds a single NE in a finite n-person non-cooperative game.

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$$EV_R = w_{RR} y_R + w_{RC} y_C \quad (5.7)$$

$$GD_R = w_{RR} y_R + w_{RC} y_C - {y_R}^{w_{RR}} {y_C}^{w_{RC}} \quad (5.8)$$

In Appendix A1, we generate a G-GD equilibrium solution for the asymmetric matching pennies game of Table 3.1(b) assuming all players share the same weights and parameters, i.e., $z_{R,a} = z_{C,a} = z_a$, $z_{R,d} = z_{C,d} = z_d$, $w_{RR} = w_{RC} = w_{CR} = w_{CC} = 0.5$. In accordance with the NE, the best response for row is to choose A' or B' at random with the choice probability at which player C is indifferent between their choices, A and B. Fig. 5.1 plots C's expected payoff from strategies A and B against the probability of R plays A'; so that the NE is the intersection of the two lines, at which R plays A' with probability 0.5. The G-GD utility transforms the payoff matrix and shifts the endpoints of these lines, relocating the equilibrium. Fig. 5.2 shows that at $z_a = 1.513$ and $z_d = 0.185$, R plays A' with probability 0.96, matching the observed choice percentages.

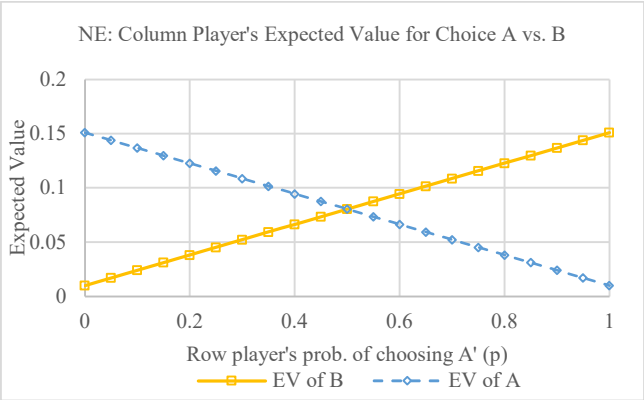


Fig. 5.1: The NE of p=0.5 at the intersection of the two lines for column player's expected payoff of A vs. B. Vertical axis is C's expected payoff; horizontal axis is the probability that R plays A'

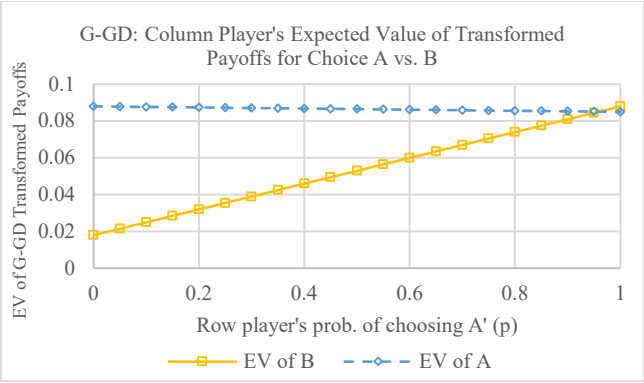


Fig. 5.2: The G-GD equilibrium when $z_a = 1.513$ and $z_d = 0.185$, using default weights, $w_{ki} = 1/n = 0.5$, results in p=0.96 at the intersection.

Figure 5.3 shows the non-linear contours of the row player's utility u_R under G-GD. For positive z_a and z_d, the adjustment term in (5.6) penalizes the utility when the player has an advantage over the other player or boosts it when the player is at a disadvantage, so u_R increases in both payoffs, but more in y_R than y_C. Above the diagonal, where $y_R < y_C$, u_R is mainly increasing in the player's own payoff, y_R. Below the diagonal, or when $y_R > y_C$, u_R is also increasing in C's payoff, but to a larger extent when moving away from the diagonal to the lower right corner. This creates a preference for points closer to the diagonal, and this creates the concave shape of the utility as displayed in Fig. 5.3.

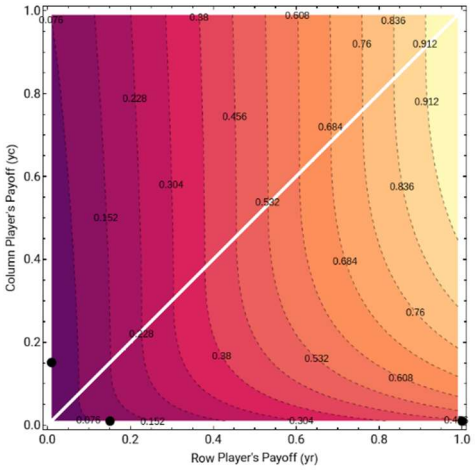


Fig. 5.3: Nonlinear G-GD contours of row player when $z_a = 1.513$, $z_d = 0.185$, and $w_{ki} = 1/n = 0.5$ over axes y_R, y_C. Note that NE contours would be linear lines parallel to the y_C axis.

E. Five Types of Behavioral Preferences in G-GD

Since $GD_k \geq 0$ (Section V.C), the sign of each parameter determines whether the player gains or loses utility from payoff dispersion in that domain. Following terminology of Fehr & Schmidt [18], Charness & Rabin [19], and Levine [43], the four sign combinations of z_a and z_d define four types of other-regarding preferences, and setting both to zero gives NE. Table 5.1 presents the five main behavioral types; the note to Table 5.1 covers the one-sided settings in which exactly one parameter is zero.

In Table 5.1, the four other-regarding behaviors (Type 2-5) correspond to the four sign combinations of nonzero z_a and z_d. One-sided behaviors occur at the intersection of two types when exactly one parameter is zero; in these cases, the player is self-interested in the domain associated with the parameter set to zero, while in the other domain, the behavior is shared by both types. For example, $z_a > 0$ and $z_d = 0$ combines the advantaged behavior common to Types 2 and 3 with self-interest when disadvantaged.

Table 5.1: The five other-regarding behavioral types for G-GD determined by the signs of z_a and z_d. The behavior column has the general n-player explanation while the last column gives the interpretation for two players.

Type	Preferences	z_a	z_d	Behavior (n-player)	2-player interpretation
1	Pure self-interest (Selfish, NE)	0	0	Payoff matrix unchanged; equilibrium is the NE	Both self-interested
2	Altruism (Social welfare)	>0	>0	Utility falls with dispersion when ahead, rises when behind	Utility increases in the opponent's payoff in both domains
3	Inequity-Aversion (Fairness, Moderate-Seeking)	>0	<0	Dispersion penalized in both domains; equal outcomes preferred	Utility increases in the opponent's payoff when ahead, decreases when behind
4	Inequity-Seeking	<0	>0	Dispersion rewarded in both domains;	Utility decreases in the opponent's payoff when ahead,

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	(Diversion-Seeking)			unequal (extreme) outcomes preferred	increases when behind
5	Competitive (Envy, Spite, Anti-social)	<0	<0	Utility rises with dispersion when ahead, falls when behind	Utility decreases in the opponent's payoff in both domains

F. Assessing the Parameters of G-GD for Real Applications

Practitioners may ask how to apply the model in real-world one-shot decision scenarios with no historical records of players' preferences. In this case, we suggest each player approximate the parameters of all players (w_{ki}, $z_{k,a}$ and $z_{k,d}$). When no prior information is available about opponents, equal parameters for all players can serve as a starting point. Then, assess which of the preference types of Section V.E is applicable and use typical default values, Section VII.C reports the types estimated for the three game categories studied in this paper. Furthermore, practitioners should consider conducting sensitivity analysis across reasonable parameter ranges to understand how parameter uncertainty affects strategic recommendations. Based on this, practitioners can simulate possible parameter settings, refine them through iterative learning from the simulated values, and finally implement them. Once the parameter estimates are determined, apply the G-GD model to find the player's best strategic decision.

VI. Real Human Behavior: Out-of-Sample Experimental Comparison of Five Models

A. Maximum Likelihood Estimation for Finding Optimal Model Parameters

To estimate the model parameters, we find the values that maximize the log-likelihood function.[9] The log-likelihood represents the probability that the model's predicted probabilities match the observed experimental probabilities using a logarithmic scale. The log-likelihood values are always negative; therefore, the best match to the experimental data occurs when the probability approaches one while its logarithm approaches zero.

Using the notation of Section II.A, the transformed payoffs ($y^*_1,\ldots,y^*_n$) for each state **s** yield a predicted set of probabilities associated $p_{(pred)k}(s_k)$ for every player k, for $k=1,2,\ldots n$, and strategy, $s_k \in \{1,2,\ldots,S_k\}$. Note that these predicted probabilities depend on the model's parameters. For each predicted probability for player k and strategy s_k, there is a corresponding observed probability, $p_{(obs)k}(s_k)$, obtained from the experimental data as the aggregated fraction of subjects who made that choice. For each player, the probabilities of choice for all strategies for that player will sum to one (whether observed or predicted).

Assessing the utility parameters involves minimizing the error between the predicted and the observed probabilities. The error minimized is the negative of the log-likelihood function. For a single player k, the log-likelihood, $\log(L_k(\boldsymbol{\theta}))$, given a model with parameter set, $\boldsymbol{\theta}$, as shown in (6.1).[10] .

$$\log(L_k(\boldsymbol{\theta})) = 100 \sum_{j=1}^{S_k} p_{(obs)k}(s_k) \log(p_{(pred)k}(s_k)) \quad (6.1)$$

This process is then repeated for every player in the game. For an entire dataset of game decisions, the log-likelihood on the whole set is generated by adding $\log(L_k(\boldsymbol{\theta}))$ for every player k and for every game in the dataset. Maximization of the total log-likelihood generates model parameters that result in the utility's best estimate of the observed probabilities on the set. [11]

B. The Five Models Tested in the Analysis

In comparative analysis, we use each model's utility to generate transformed payoff matrices and equilibrium predictions to compare their ability to predict subjects' preferences from experimental data. Table 6.1 presents the models included in this paper's comparison, along with their mathematical equations and parametric properties. The model is represented as a utility function that transforms each state, represented by the cell (y_R, y_C) in the m×k payoff matrix, where the row player has m strategies, and the column player has k strategies.[12] NE is used as the baseline for the comparison. As the table shows, G-GD introduces nonlinearity in payoffs relative to NE, FS, and CR[13], while G-GD's nonlinearity allows for additional behavioral representations relative to BO.

Table 6.1: List of the five models compared in the experimental analysis, each with parameter count, parameter limitations, and linearity property in payoffs. NE is used as a benchmark.

Model	Model Description	Linear in y	No. of Para.	Parameters and their Limits
NE	$u_R(y_R, y_C) = y_R$	Yes	0	N/A
FS	$u_R(y_R, y_C) = \begin{cases} y_R + \sigma(y_C - y_R), & \text{if } y_R \le y_C \\ y_R - \rho(y_R - y_C), & \text{if } y_R \ge y_C \end{cases}$	Yes	2	$\sigma < 0$ $0 < \rho \le 1$
CR	$u_R(y_R, y_C) = \begin{cases} y_R + \sigma(y_C - y_R), & \text{if } y_R \le y_C \\ y_R - \rho(y_R - y_C), & \text{if } y_R \ge y_C \end{cases}$	Yes	2	σ unrestricted in sign $0 < \rho \le 1$
BO	$u_R(y_R, y_C) = y_R - \sigma\left(\frac{y_C - y_R}{y_C + y_R}\right)^2$	No	1	$\sigma > 0$
G-GD	$u_R(y_R, y_C) = \begin{cases} y_R + z_d GD_R, & \text{if } y_R \le y_C \\ y_R - z_a GD_R, & \text{if } y_R \ge y_C \end{cases}$ where GD_R as in (5.8) with $w_{R,R} = w_{R,C} = 0.5$	No	2	z_a, z_d unrestricted in sign

[9] We implement this estimation using the MATLAB Optimization toolbox [56], a NE solver, *npg* [50], and a custom MATLAB function *lolik2xn_or_m_oos_init* that applies each model's transformations to the payoff matrix and estimates the parameters by maximizing log-likelihood. The implementation code & detailed results are in the journal supplement [34].

[10] The constant multiplier of 100 represents the likelihood of observations from 100 subjects to prevent log-likelihood values from getting too small.

[11] In experiments, aggregated choice percentages of the same homogeneous DMs were used for both row and column players and across different games in the same dataset, we used the same parameters of each model across all games and players.

[12] In G-GD, the payoffs are normalized to (0,1] before the transformation is applied, using a common minimum and maximum computed across each dataset, whose games were verified to share a payoff range. The resulting utilities are then rescaled back to the original payoff range before the NE solver is invoked. This keeps the parameters z_a and z_d independent of the scale of the input payoffs.

[13] While FS and CR reduce to the same functional form in 2-player games, the parameter constraints are different as presented in Table 6.1. FS (or Fehr & Schmidt's fairness utility [14]) is designed to represent only the inequity aversion behavioral case ($\sigma < 0$), where the players aim to achieve equal payoffs. CR (or Charness & Rabin's social welfare utility [19]) not only can depict the inequity aversion case ($\sigma < 0$) but also represents the altruism behavioral case ($\sigma > 0$). Due to these different constraints and the out-of-sample experimental test, comparing FS and CR separately is necessary since the CR model may underperform FS due to overfitting.

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Each of the five models nests NE as a special case by adding one design element to NE. FS and CR add an advantaged and disadvantaged adjustment term that is linear in payoff differences, BO adds a fixed nonlinear function applied in both domains, and G-GD adds an advantaged and disadvantaged adjustment term that is nonlinear in payoffs. Therefore, the comparison in Section VI.D attributes differences in predictive performance to these specific features rather than to model complexity alone.

In the experimental analysis of the five models on the aggregate choice data, we estimate a single set of parameters for all players using aggregate-level choice frequencies from the experimental data. This is consistent with the use of aggregate subject choice frequency to compare models of game theory in [3] and [29]. Each model predicts an equilibrium distribution over strategies, and an individual subject's single choice in a given game is a draw from that distribution. This assumption of parameter homogeneity reflects the extent to which an equilibrium prediction is observed, rather than a claim about individual-level heterogeneity in social preferences.

C. Experimental Data Sets & Out-of-Sample Comparisons

The experimental data included in the analysis of this paper consists of real subject choices in games, which are sourced from well-cited, publicly available incentivized experimental studies that cover a wide variety of games. From the experimental sources, we compiled three datasets (Datasets 1, 2, and 3) of games with similar characteristics for out-of-sample testing of the models. Out-of-sample analysis is frequently used in economics and decision theory to compare decision-making models ([29], [44], [27], [30]) because it puts models with different numbers of parameters on equal footing. Table 6.2 summarizes the experimental datasets, including sources, study characteristics, game types, payoff-matrix sizes, numbers of data points, and in-sample set sizes in each dataset.

For each experimental data source, we tabulated the percentage of players choosing each strategy, aggregated across all subjects' choices in each game. For each game, we treat a player's full choice distribution as a single data point the model aims to predict. For symmetric games in which the row and column players share the same payoff matrix and strategies, the data for both players is combined into one aggregate percentage distribution. For each dataset in Table 6.2, we randomly select approximately 80% of the data points for the in-sample set and use the remaining 20% as the out-of-sample set. We repeat this random selection eight times per dataset and refer to each as a repetition. Within each repetition, we estimate model parameters only on the in-sample set and evaluate predictive performance on the out-of-sample set. This tests whether a model's predictive advantage holds when the data changes and does not reward model complexity. An additional parameter that fits noise in the in-sample set will be detrimental to a model's out-of-sample evaluation.

Each dataset contains observations from games with different characteristics. Dataset 1 in Table 6.2 includes three types of simple strategic games with asymmetric payoffs from [29][14]. Dataset 2 consists of symmetric games with varied dominance, including strictly and weakly dominance-solvable games and games with unique pure, unique mixed, and multiple equilibria, and this dataset comprises 10 games from [45] and 12 similar games from [46]; the five games from [46] with different properties were excluded. Dataset 3 comprises 28 choice distributions in complex games from [47] that are described as cognitively challenging since the dominance, if any, is not obvious and may require several dominance eliminations to detect.[15] All data is from real human decisions in studies of experimental behavior in games and includes 80 data points across various sources.

Table 6.2: Properties of the three datasets used in the out-of-sample experimental comparisons.

	Source of behavioral data	Characteristics of the Study	Types of Games	Size of payoff matrix	No. of data points	In-sample set size
Dataset 1	Kulpmann & Kuzmics [29]	147-166 subjects/game, 6260 total observations	Hawk-Dove, Rock-Paper-Scissors, Matching Pennies	2x2, 3x3	30	24
Dataset 2	Rogers, Palfrey, Camerer [46]	74 subjects, 1184 total observations	Symmetric with varied dominance	3 x 3	22	18
	Stahl & Wilson [45]	40 subjects, 400 total observations				
Dataset 3	Costa-Gomes, Crawford, Broseta [47]	92 subjects, 1656 total observations	Complex games with mixed strategic structures [a]	3x2, 2x3, 2x4, 4x2.	28	22

[a] dominance, iterated dominance, pure equilibria w/o dominance

D. Results of the Out-of-Sample Model Comparison on Predicting Behavior in Experimental Data

Table 6.3: Out-of-sample LL for each model prediction averaged over 8 repetitions (with standard deviation in parentheses). G-GD has the best LL (highest). *NE is used as a benchmark.

	NE*	FS	CR	BO	G-GD
Dataset 1, mean (SD)	-504.58 (43.00)	-516.55 (60.15)	-488.24 (57.21)	-504.58 (43.00)	-466.10 (46.71)
Dataset 2, mean (SD)	-2784.48 (1093.37)	-2461.68 (940.24)	-2324.46 (559.24)	-3006.14 (718.13)	-1360.98 (699.75)
Dataset 3, mean (SD)	-4483.17 (1050.44)	-2697.51 (953.71)	-2597.73 (745.06)	-1999.39 (887.59)	-1371.81 (632.62)

In this section, we show the results of a model comparison following an out-of-sample testing method with eight repetitions. Section VI.B presented the models compared. Table 6.2 in Section VI.C details the three datasets used in this paper's comparative analysis. Table 6.3 presents the summary results with a row for each dataset, showing the log-likelihood (LL) of each model prediction on the out-of-sample set, averaged over the eight repetitions for that dataset, followed by the standard deviation (SD) of repetitions in parentheses. As the table shows, G-GD achieves the best prediction (i.e., maximum log-likelihood) across all five models on all three datasets, with lower standard deviations across the eight repetitions than the closest competing model: CR in Dataset 1 and BO in Dataset 3.

[14] The ten Hawk-Middle-Dove games in data of [29] excluded from Dataset 1 of Table 6.2 have the same payoffs for row and column players. In this case, all of the compared preference models reduce to NE.

[15] Four 2x2 games in [47] have clear equilibria because of the small size; therefore, they did not share the same characteristics as the larger problems in being cognitively challenging; therefore, these were excluded from Dataset 3.

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In Dataset 2, G-GD's mean LL exceeds that of the next-best model, CR, by more than one standard deviation. Table 6.4 provides additional insights into the eight individual repetitions; it shows how many times out of eight each model's prediction has the best log-likelihood among all compared models in each dataset. The last row has the overall count across all datasets, which shows that the G-GD model predicts best in 21 of the 24 out-of-sample LL comparisons across all three datasets. Further detailed results for each of the eight repetitions can be found in the online supplement [34].

Table 6.4: The number of repetitions for which each model has the best out-of-sample prediction LL out of eight repetitions per dataset. G-GD is best per dataset and overall. *NE is used as a benchmark.

No. of wins per model in out-of-sample LL	NE*	FS	CR	BO	G-GD
Dataset 1	0	0	0	0	8
Dataset 2	0	0	1	0	7
Dataset 3	0	1	0	1	6
Total wins (out of 24)	0	1	1	1	21

Table 6.5 presents the Vuong statistical likelihood-ratio test for non-nested models [48] comparing G-GD to its closest competing models, CR and BO, in each dataset. The Vuong statistic is computed on N log-likelihood differences to test whether they differ significantly from zero. N is the pool of individual out-of-sample data points for these models across the eight repetitions (N = 48 for Datasets 1 and 3; N = 32 for Dataset 2). For each comparison, the table reports the mean and standard deviation of the paired log-likelihood differences, the Vuong test statistic (z-score of the paired log-likelihood differences, positive favors G-GD), its associated p-value, and a Holm-Bonferroni-adjusted p-value controlling the family-wise error rate across the six comparisons [49]. Across all rows of Table 6.5, both the raw and Holm-adjusted Vuong p-values (the last 2 columns) fall below 0.05 in every comparison, indicating that G-GD's advantage in predictive fit is unlikely to be due to chance even after accounting for the number of comparisons.

Table 6.5: Vuong statistical likelihood-ratio test comparing G-GD's out-of-sample LL to that of CR and BO in three datasets across all eight repetitions, with raw and Holm-Bonferroni-adjusted p-values.

Out-of-sample Comparison	N	Mean of LL Diff.	SD of LL Diff.	Vuong statistic	Vuong p	Holm-Bonferroni Adjusted Vuong p
Dataset 1: G-GD vs. CR	48	3.69	7.19	3.56	0.0004	0.0011
Dataset 1: G-GD vs. BO	48	6.41	7.48	5.94	0.0000	0.0000
Dataset 2: G-GD vs. CR	32	240.87	569.82	2.39	0.0168	0.0309
Dataset 2: G-GD vs. BO	32	411.29	598.95	3.88	0.0001	0.0005
Dataset 3: G-GD vs. CR	48	204.32	382.03	3.71	0.0002	0.0008
Dataset 3: G-GD vs. BO	48	104.60	299.26	2.42	0.0155	0.0309

Note: N is the number of out-of-sample data points over which LL of the models are compared. Reported next are the mean and standard deviation of the difference in LL values over N comparisons. $p<0.05$ in the last column shows G-GD's advantage is statistically significant across all three datasets.

Table 6.6 presents G-GD's estimated parameters for each dataset and the behavior type they represent, with the range of values across the eight repetitions. Since the parameters are estimated on the in-sample set only and then tested in the out-of-sample set, overfitting would lower the out-of-sample log-likelihood rather than raise it (Section VI.C). Table 6.6 also shows that the signs of z_a and z_d are the same across all eight repetitions for each dataset, despite greater variation in z_d for Dataset 2 and Dataset 3. This sign stability, together with G-GD's out-of-sample performance, supports reading the behavior types, though not the exact parameter values, as descriptions of aggregate subject behavior rather than as fitting constants. Section VII.C discusses how these types differ by game category.

Table 6.6: Estimated parameters for G-GD on each dataset (mean across the eight repetitions and range of values in brackets) followed by the other-regarding behavior type they show (Section V.E). Assessed parameters have consistent signs within each dataset.

	Nature of Games	Mean z_a [range]	Mean z_d [range]	Behavior Type	Parameter Condition
Dataset 1	Simple strategic games with asymmetric payoffs	1.9 [1.9, 2.0]	0.1 [0.1,0.1]	Type 2: Altruism	$z_a > 0$ $z_d > 0$
Dataset 2	Symmetric with varied dominance	-8.2 [-10.0, -4.2]	10.4 [2.6, 24.9]	Type 4: Inequity-Seeking	$z_a < 0$ $z_d > 0$
Dataset 3	Complex strategic games with mixed structures[a]	11.8 [9.3, 16.3]	-9.1 [-15.9, -4.7]	Type 3: Inequity-Aversion	$z_a > 0$ $z_d < 0$

[a] dominance, iterated dominance, pure equilibria w/o dominance

VII. Discussion and Limitations

A. Experimental Verification of G-GD

Across the three datasets, G-GD attains the best out-of-sample log-likelihood among the five models compared (Table 6.3), and its advantage over the closest competitors (i.e., BO and CR models) is statistically significant (Table 6.5). In light of these results, we reflect back on the failures of other models documented in Section V.A. In mean out-of-sample log-likelihood, BO is the strongest competitor in Dataset 3, but the weakest model in Dataset 2, below NE, and the order of FS and BO reverses from one dataset to the next (Table 6.3), mirroring the finding of Engelmann and Strobel [21] that FS's advantage over BO depends on the setting. In addition, Section V.D shows that G-GD reproduces the asymmetric matching pennies behavior that Boylan and Grant [12] show linear inequity aversion cannot.

The results do not imply that payoff nonlinearity alone accounts for departures from Nash equilibrium, since other factors may also contribute. What the comparison shows is that nonlinear other-regarding preferences predict better than the linear and fixed-nonlinear alternatives tested.

B. Computational Complexity & Scalability

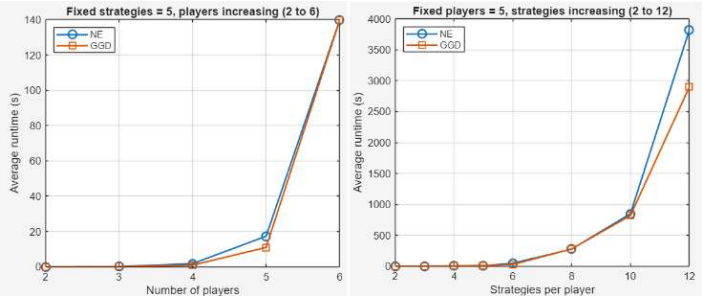
This section compares the computational cost of G-GD with that of NE, the benchmark, as the game grows in players and strategies. While both methods use the code, *npg* [50], for the equilibrium calculation, G-GD also uses the non-linear payoff transformation built on top of it. Computing the NE even in two-player games is PPAD-complete [51], and *npg* is not of polynomial order [36]. The G-GD transformation visits each entry of the payoff matrix a fixed number of times, so its

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computational cost is linear in the matrix size.

Fig. 7.1 graphs the average runtime for G-GD and NE in solving games with randomly generated payoff matrices in the range [0,1] over five trials. While larger games take significantly longer to solve for both NE and G-GD, Fig. 7.1 shows that G-GD does not add substantial computational overhead in either scalability test case, and neither method is consistently faster.

When an exact NE is impractical, distributed seeking algorithms [4] [5] [6] [7] [8] approach the equilibrium iteratively. Table 7.1 presents the complexity order in an n-player m-strategy game for an exact NE solver, a distributed approximation, the G-GD transformation for each, and the overhead ratio. In both cases, the G-GD transformation does not change the order of growth. Fig. 7.2 illustrates these, with the paired curves for each method staying together as the problem size grows. Additional details of this section's test setup and analysis are in supplement [34].



(a) (b)

Fig. 7.1: Average runtime for G-GD and NE over 5 trials: (a) Strategies per player fixed at 5, players increasing. (b) Players fixed at 5, strategies per player increasing. (MATLAB R2025b on an Intel Core Ultra 7, 2.2 GHz, Windows 11)

Table 7.1: Complexity order for an NE solver and a distributed approximation with overhead from the G-GD transformation, n-player m-strategy game with m^n strategy profiles, $n \cdot m^n$ payoff matrix.

	Order	GGD transform	Overhead ratio
npg NE solver [36] [50] (based on measurements)	$O(m^{1.3n})$	$O(n \cdot m^n)$	$n \cdot m^{-0.3n}$
Distributed-Seeking [4]- [8] I iterations q neighborhood, where q<n	$O(I \cdot n \cdot m)$	$O(I \cdot n \cdot q)$	q/m

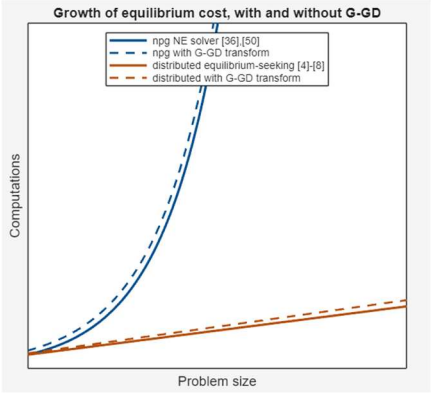


Fig. 7.2. Schematic comparison of the increase of equilibrium computation cost with problem size. Axes are in arbitrary units graphed based on the complexity order of algorithms. Solid curves are NE computation, dashed are with the G-GD transformation.

The parameters are assessed once before the game is solved (Sections V.F, VI.A and VII.C), so obtaining them does not add to the cost of solving for the equilibrium.

C. Other-Regarding Behavior Types by Game Category

Table 6.6 shows that the preference type estimated by G-GD differs by game category, with parameter signs that are consistent in all eight repetitions. Within the three categories studied, we propose the following default behavior types:

- **Simple strategic games with asymmetric payoffs** (e.g., Dataset 1): altruism ($z_a > 0$, $z_d > 0$). The clear payoff differences of simple asymmetric games may make it easy for players to see how their choice affects the opponent, and to favor outcomes that also benefit the opponent.
- **Symmetric games with varied dominance** (e.g., Dataset 2): inequity-seeking ($z_a < 0$, $z_d > 0$). These parameter settings give every unequal outcome (off-diagonal cells) a positive dispersion adjustment and the equal outcomes on the diagonal none.
- **Complex strategic games** (e.g., Dataset 3): inequity aversion ($z_a > 0$, $z_d < 0$). Since dominance is not obvious and requires high cognitive effort, players may fall back on avoiding unequal outcomes as a simpler rule for choosing.

These guidelines describe the qualitative behavior type that each category falls into, not fixed numeric values for the parameters. This context dependence is consistent with survey evidence that recognition of dilemma intensity varies across strategic settings [52]. The z_a, z_d values in Table 6.6 are point estimates from the datasets studied and should be treated as a starting reference rather than universal defaults.

D. Limitations and Future Directions

The model is tested on two-player games, and the n-player formulation in Section V.C has not yet been evaluated against multi-player data. The two-player setting isolates these strategic behaviors without confounding them with equilibrium selection and belief formation, which multi-player generalizations of these games typically introduce. In addition, the use of two-player games follows standard practice in behavioral game theory, where mechanisms are first isolated in simple settings before exploring more complex environments [29].

The models in this study predict the share of subjects choosing each strategy, and the data we compare against are these aggregate choice frequencies, not individual decisions. Model parameters are therefore estimated at the aggregate level for all the models in the comparison. The estimated parameters, and the preference types inferred from them, describe representative-agent behavior and do not establish whether individual subjects differ in their social preferences. A finite-mixture extension allowing K subtypes would introduce $3K-1$ parameters against in-sample sets of 18 to 24 observations, which the out-of-sample validation design of Section VI.C would likely penalize. Separating heterogeneity from representative-agent behavior calls for individual-level data in

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which each subject plays many one-shot games, so that individual behavior can be identified from within-subject consistency; we leave this as a subject for future research.

The experimental behaviors reported in Section VII.C are derived from the three game categories represented in Datasets 1 through 3. This does not preclude the use of G-GD in generalized decision contexts (see Section V.F), but its predictive value outside them has not been tested. Future research may test how far behavior generalizes across different game categories. A comparison of G-GD with other models on the prisoner's dilemma is the subject of our ongoing work.

Beyond these limitations, the G-GD other-regarding preferences developed here point to two lines of extension: equitable resource allocation in multi-agent and distributed settings [5], [7], [10], [53], [54] or as spatial and evolutionary game-theoretic models of cooperation [55].

VIII. Conclusions

NE assumes that each player cares only about their own payoff, yet experiments show that choices in one-shot games also respond to changes in the opponents' payoffs, in ways that risk, noise, or bounded rationality alone cannot explain [11] [12] [13]. In this paper, we propose G-GD, a nonlinear other-regarding model in which the player's utility responds differently to being ahead of or behind the opponents. On behavioral experimental data in two-player, one-shot games, G-GD attains the highest mean out-of-sample log-likelihood of the five models compared on all three datasets tested (Table 6.3). G-GD predicts best in 21 of the 24 total repetitions (Table 6.4), and Vuong tests against its closest competitors, CR and BO, remain significant at 5% level in every dataset after Holm-Bonferroni adjustment (Table 6.5). G-GD's advantage comes from applying GD [37], a hybrid arithmetic-geometric function of the players' payoffs, separately in the advantaged and disadvantaged branches using only two structural parameters (the same number as in the FS and CR models), and no auxiliary assumptions or free parameters. We also show that the G-GD nonlinear payoff transformation adds negligible computational cost to solving for the equilibrium (Section VII.B).

The relation of the two parameters of G-GD distinguishes five behavioral types (Section V.E): self-interest, which corresponds to NE, and four other-regarding preferences, namely altruism, inequity aversion, inequity-seeking, and competitive preferences. Out of the other-regarding models compared (Table 6.1), FS and BO models can only depict inequity-aversion, while CR model can represent all types except inequity-seeking. The experimental analysis showed that the G-GD behavior type identified at the aggregate level differed by game category, and that the signs of the parameters are consistent across all eight repetitions within each dataset (Table 6.6). In none of the three datasets do the estimates fall into Type 1, the NE case, in any repetition, which is consistent with NE never being the best predictor (Table 6.4). The estimates for the simple asymmetric games of Dataset 1 fall into Type 2 (altruism), which FS and BO cannot represent. The complex games of Dataset 3 fall into Type 3 (inequity aversion), which FS, CR, and BO can all represent, but G-GD still predicts best among these. In the symmetric games of Dataset 2, the estimates fall into Type 4 (inequity-seeking), which lies outside the ranges of FS, CR, and BO.

Appendix

A1 Solving the Asymmetric Matching Pennies Game by G-GD

Tables A1.1 and A1.2 give the normalized and G-GD-transformed payoff matrices of the asymmetric matching pennies game of Table 3.1(b), respectively, from which the equilibria in Figs. 5.1 and 5.2 are computed. The NE solution for the row player is $p_R(A') = 0.5$ and $p_R(B') = 0.5$ and for column player is $p_C(A) = 0.125$ and $p_C(B) = 0.875$. On the other hand, using the G-GD utility with parameters $z_a = 1.513$, $z_d = 0.185$ for the game in Table A1.2 results in the equilibrium solution with predicted probabilities, $p_R(A') = 0.96$, $p_R(B') = 0.04$, $p_C(A) = 0.16$, and $p_C(B) = 0.84$, which are a perfect match to experimental observations.

Table A1.1: Normalized payoffs of asymmetric matching pennies; observed experimental choices are shown in parentheses next to each strategy

		Column Player (C)			
		A (16%)		B (84%)	
Row Player (R)	A' (96%)	1	0.01	0.01	0.151
	B' (4%)	0.01	0.151	0.151	0.01

Table A1.2: G-GD payoff matrix transformed from Table A1.1, with optimized parameters $z_a = 1.513$, $z_d = 0.185$.

		Column Player (C)			
		A (16%)		B (84%)	
Row Player (R)	A' (96%)	0.387	0.085	0.018	0.088
	B' (4%)	0.018	0.088	0.088	0.018

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ABBREVIATIONS AND NOMENCLATURES

DM	Decision Maker
Game Models	
NE	Nash Equilibrium
FS	Fehr-Schmidt fairness model
CR	Charness-Rabin social welfare model
BO	Bolton-Ockenfels inequity aversion model
G-GD	Game-Geometric Dispersion
QRE	Quantal Response Equilibrium

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CH	Cognitive Hierarchy
Level-k	Level-k Bounded rationality

GDT multi-criteria alternative notations:

n	number of criteria
Y	*alternative with associated criteria values* $(y_1, \ldots, y_n)$
j	the index of the criterion, where $j=1,2,\ldots,n$
y_j	the value of criteria j in alternative Y
w_j	weight of importance of criteria j, varies by DM
EV	Linear Value, or weighted arithmetic mean
GD	Geometric Dispersion measure
GDT	Geometric Dispersion Theory
z	the extent of concave preferences, varies by DM

n-player game notations:

n	number of players, also number of payoffs for players in state **s**
k	index of the player, where $k= 1, 2, \ldots, n$
S_k	number of finite strategies for Player k
s_k	strategy selected by player k, where $s_k=1,\ldots, S_k$
s	state, represented by the set of strategies selected by all players, i.e., **s** = $[s_1, \ldots, s_k, \ldots, s_n]$
$(y_1, \ldots, y_n)$	payoff earned by each player at state **s**
$(y^*_1, \ldots, y^*_n)$	transformed payoffs at state **s**, where $y^*_k = u_k(y_1, \ldots, y_n)$
$u_k(y_1, \ldots, y_n)$	utility of player k as a function of payoffs of all players at **s**
$p_k(s_k)$	probability for player k for selecting strategy s_k

Two-player game notations:

R	Row player
C	Column player

(y_R, y_C)	The pair of payoffs earned in each state by row and column players, respectively
$p_R(A')$	Row player's probability of choosing strategy A'
$p_C(A)$	Column player's probability of choosing strategy A
$u_R(y_R, y_C)$	utility of row player (R) as a function of payoffs (y_R, y_C)
$u_C(y_R, y_C)$	utility of column player (C) as a function of payoffs (y_R, y_C)
(y^*_R, y^*_C)	utility transformed payoff of row/column, or $(u_R(y_R, y_C), u_C(y_R, y_C))$

G-GD Notations

k	the index of the player whose utility we are deriving, or u_k
i	the index of the player's payoff in the EV or GD calculation, or $(y_1,\ldots, y_i,\ldots,y_n)$, where $i = 1,2,,\ldots, n$.
w_{ki}	player k's weights of importance of payoff, y_i
EV_k	player k's weighted mean of payoffs $(y_1,\ldots,y_n)$ in state **s**
GD_k	player k's weighted geometric dispersion measure of $(y_1,\ldots,y_n)$
$z_{k,a}$	advantaged coefficient for player k
$z_{k,d}$	disadvantaged coefficient for player k

Note: When it is assumed that all players have the same weights, w_{ki}, $z_{k,a}$, and $z_{k,d}$, all above may appear without the k subscript, or w_i, z_a, z_d, EV, and GD.

Maximum Log-Likelihood Estimation

$\boldsymbol{\theta}$	a vector of model parameters, not specific to a model
$L_k(\boldsymbol{\theta})$	Likelihood function for player k
$log(L_k(\boldsymbol{\theta}))$	Log-likelihood function for player k
$p_{(obs)k}(s_k)$	observed probability (in experiments) for player k's strategy s_k
$p_{(pred)k}(s_k)$	predicted probability (using model) for player k's strategy s_k